

Jashuva Billa

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Education

Jawaharlal Nehru Technological University Hyderabad
Bachelor of Technology in Computer Science

Hyderabad, India
Aug 2019 - July 2023

Experience

Innovapath IT solutions
AI Engineer

Hyderabad, India
feb 2024 – Present

- Built enterprise Generative AI and Agentic AI assistants for multi-step customer support workflows, combining RAG, LangGraph orchestration, domain agents, tool calling, memory, and policy-driven execution
- Designed stateful LangGraph workflows with intent-based routing, conditional branching, parallel execution, reflection loops, checkpointing, retries, and resumability for long-running multi-step workflows.
- Designed multi-agent architectures with triage, knowledge/troubleshooting, service, billing, order-management, escalation, and policy agents coordinated through a LangGraph orchestrator.
- Implemented MCP-based tool execution for CRM, scheduling, billing, and diagnostic workflows with schema validation, retries, timeouts, idempotency controls, and standardized execution results.
- Built RAG and retrieval components using Milvus for semantic search, BM25 for keyword retrieval, graph-based retrieval, Reciprocal Rank Fusion (RRF), and cross-encoder reranking for grounded context selection.
- Implemented context engineering and memory management using conversation history, session state, persistent memory, context compression, context filtering, Redis/ElastiCache caching, and dynamic context selection.
- Built end-to-end RAG pipelines from document ingestion and preprocessing through metadata extraction, structure-aware chunking, embeddings, query cleaning, multi-query expansion, retrieval, reranking, grounded generation, and source citation.
- Implemented hybrid retrieval combining dense semantic search, BM25 keyword search, and graph retrieval; used RRF, Top-K tuning, embedding similarity thresholds, keyword validation, and cross-encoder scores to improve retrieval relevance.
- Built retrieval quality and fallback logic using cross-encoder reranking, irrelevant-result detection, similarity thresholds, keyword validation, confidence scoring, and retrieval fallbacks when sufficient grounding was unavailable.
- Implemented AI safety and governance using PII/PHI detection and redaction, Bedrock Guardrails, RBAC, policy validation, sensitive-action gating, human-in-the-loop approval, and audit logging.

- Built evaluation pipelines for retrieval, generation, safety, and agent behavior using RAGAS, DeepEval, LLM-as-a-Judge, Precision@K, Recall@K, nDCG, BERTScore, ROUGE, hallucination checks, tool correctness, and task-completion metrics.
- Optimized RAG and agent latency using query-embedding caching, retrieval-result caching, prompt caching, Redis/ElastiCache, streaming responses, and efficient retrieval pipelines.
- Implemented LLM and agent observability using Langfuse, OpenTelemetry, Prometheus, Grafana, and CloudWatch to trace prompts, retrieval, tool calls, agent workflows, latency, failures, and system health.
- Deployed containerized AI services on AWS using EKS, ECR, Bedrock, API Gateway, Lambda, S3, CloudWatch, IAM, Secrets Manager, and KMS, with CI/CD pipelines for automated deployment and operational reliability.
- Built Python/FastAPI backend services for chat, workflow orchestration, tool execution, health checks, evaluation endpoints, and integration with frontend and enterprise systems.
- Designed failure-handling mechanisms including bounded retries, timeouts, fallback and escalation paths, idempotent actions, health checks, checkpointing, and resumable workflows.

Technical Skills

Programming: Python, SQL

Generative AI & LLMs: Generative AI, LLMs, RAG, Agentic AI, Multi-Agent Systems, ReAct, Function Calling, Prompt Engineering

Agent Frameworks: LangGraph, LangChain, MCP

RAG & Retrieval: Milvus, Semantic Search, BM25, Hybrid Retrieval, Graph Retrieval, Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, Top-K Retrieval

Document Processing: Docling, Metadata Extraction, Structure-Aware Chunking, OCR

Embeddings: Sentence Transformers, all-MiniLM-L6-v2, E5

Evaluation: RAGAS, DeepEval, LLM-as-a-Judge, Precision@K, Recall@K, nDCG, F1, BERTScore, ROUGE

Guardrails & Security: Bedrock Guardrails, Presidio, PII/PHI Redaction, RBAC, IAM, Policy Validation, Human-in-the-Loop

Backend & Data: FastAPI, REST APIs, PostgreSQL, Redis/ElastiCache, Milvus

Cloud & DevOps: AWS Bedrock, EKS, ECR, Lambda, API Gateway, S3, SageMaker, CloudWatch, Docker, Kubernetes, CI/CD

Observability: Langfuse, OpenTelemetry, Prometheus, Grafana, CloudWatch